

HORNSBY GIRLS' HIGH SCHOOL



2007 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 2

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using a black or blue pen
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown for every question.
- Begin each question on a fresh sheet of paper.

Total marks (120)

- Attempt Questions 1-8
- All questions are of equal value

Total Marks – 120

Attempt Questions 1-8

All Questions are of equal value

Begin each question on a NEW SHEET of paper, writing your student number and question number at the top of the page. Extra paper is available.

Question 1 (15 marks) Use a SEPARATE sheet of paper. **Marks**

(a) Use the technique of integration by parts to find:

(i) $\int \ln x \, dx$ **2**

(ii) $\int e^x \cos x \, dx$ **3**

(b) Use partial fractions to find $\int \frac{4dx}{4x^2 - 1}$ **2**

(c) Find $\int \frac{dx}{x^2 + 2x + 4}$ **2**

(d) Find $\int \sqrt{\frac{x-1}{x+1}} \, dx$ **2**

(e) By using the substitution $t = \tan\left(\frac{x}{2}\right)$ and partial fractions evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{4 \sin x + 3 \cos x}$ **4**

Question 2 (15 marks) Use a SEPARATE sheet of paper. **Marks**

(a) Given that P and Q represent the complex numbers $5 + 2\sqrt{6}i$ and $1 - \sqrt{3}i$ respectively, find:

(i) $\frac{P}{Q}$ in the form $x + iy$ **2**

(ii) $\overline{P} \times \overline{Q}$ **2**

(iii) $\sqrt{P}$ in the form $x + iy$ **2**

(iv) The modulus and argument of Q **2**

(v) The complex number R in the form $x + iy$, given that $\arg R = 2 \arg Q$
and $|R| = 2|Q|$ **2**

(b) On an Argand diagram sketch the region defined by $-2 \leq \operatorname{Re}(Z) < 1$ **1**

(c) Draw a sketch in the complex plane of the locus of Z given by the equations

(i) $\arg(Z - 3 + 2i) = \frac{\pi}{4}$ **2**

(ii) $\arg(Z - 1) - \arg(Z + 1) = \frac{\pi}{2}$ **2**

Question 3 (15 marks) Use a SEPARATE sheet of paper. **Marks**

- (a) Given $f(x) = e^x - 2$ draw large (half page), separate, neat and accurate sketches of each of the following, showing clearly all the intercepts and asymptotes:
- (i) $y = f(x)$ 2
 - (ii) $y = |f(x)|$ 2
 - (iii) $y = \frac{1}{f(x)}$ 2
 - (iv) $y^2 = f(x)$ 2
- (b) The region bounded by the curve $y = x^2 - 4x + 4$ and the x and y axes is rotated about the line $y = -1$. Find the volume of the solid of revolution. 4
- (c) An ellipse has equation $\frac{x^2}{4} + \frac{y^2}{9} = 1$. Find the eccentricity, co-ordinates of the foci S and S' and the equations of the directrices. 3

Question 4 (15 marks) Use a SEPARATE sheet of paper. **Marks**

- (a) Find:
- (i) $\int \sin^3 x \cos^5 x \, dx$ 3
 - (ii) $\int \frac{dx}{x^3 \sqrt{x^2 - 4}}$ 3
 - (iii) $\int \tan^4 x \, dx$ 3
- (b) (i) Show that a reduction formula for, $I_n = \int x^n \cos x \, dx$, is
- $$I_n = x^n \sin x + nx^{n-1} \cos x - n(n-1)I_{n-2}.$$
- 3
- (ii) Hence, or otherwise, evaluate $\int_0^{\pi/2} x^4 \cos x \, dx$ 3

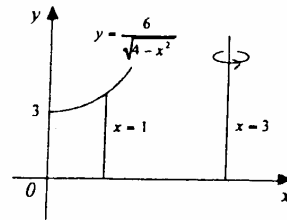
Question 5	(15 marks)	Use a SEPARATE sheet of paper.	Marks
(a)	A mass of 3 kg, on the end of a string 0.8 metres long, is rotating as a conical pendulum with angular velocity 3π radians per second. Use $g = 10\text{ m/s}^2$ and let θ be the angle that the string makes with the vertical.		
(i)	Draw a diagram showing all the forces acting on the mass		1
(ii)	By resolving forces, find the tension in the string		2
(iii)	Find θ correct to the nearest degree		1
(b)	A particle is dropped from rest at a height h metres above the ground. At time t seconds its height above the ground is given by		
	$x = h + \frac{gt}{k} + \frac{ge^{-kt}}{k^2} - \frac{g}{k^2}$		
(i)	Show that $\ddot{x} = g - kv$ where the velocity of the particle is v m/s		2
(ii)	What forces are acting on this particle? Explain carefully.		1
(iii)	If it takes T seconds for the particle to reach half its terminal velocity, find the value of e^{kT} .		2
(c)	Find the magnitude of the braking force required to stop a truck of mass 4800 kg in 55 metres when it is traveling at 40 km/h down an incline of angle 5° to the horizontal. (assume no wind resistance and use $g = 10\text{ m/s}^2$)		3
(d)	Prove the identity $\frac{\cos y - \cos(y + 2x)}{2 \sin x} = \sin(y + x)$		3

Question 6 (15 marks)

Use a SEPARATE sheet of paper.

Marks

(a)



A mould for a section of concrete piping is made by rotating the region bounded by the curve $y = \frac{6}{\sqrt{4-x^2}}$ and the x -axis between the lines $x = 0$ and $x = 1$ through one complete revolution about the line $x = 3$. All measurements are in metres.

- (i) By considering strips of width δx parallel to the axis of rotation, show that the volume $V \text{ m}^3$ of the concrete used in the piping is given by $V = 12\pi \int_0^1 \frac{3-x}{\sqrt{4-x^2}} dx$ **3**

- (ii) Hence, or otherwise, find the volume of the concrete used in the piping, giving your answer correct to the nearest cubic metre. **3**

- (b) (i) Sketch the graph of the curve $y = x + e^{-x}$ showing clearly the coordinates of any turning points and the equations of any asymptotes. **2**

- (ii) The region in the first quadrant between the curve $y = x + e^{-x}$ and the line $y = x$ and bounded by the lines $x = 0$ and $x = 1$ is rotated through one complete revolution about the y -axis. Use the method of cylindrical shells to find the volume of the solid. **5**

- (c) The expression $\sqrt{12 + \sqrt{12 + \sqrt{12 + \sqrt{12 + \sqrt{\dots}}}}}$ has a limit L . Find the exact value of L . **2**

Question 7 (15 marks) Use a SEPARATE sheet of paper. **Marks**

(a) The roots of $px^3 + qx^2 + rx + s = 0$ form a geometric series. Prove that $pr^3 = q^3s$ **3**

(b) If i is a root of $z^4 + 2z^3 - 2z^2 + 2z - 3 = 0$, find the other three roots. **3**

(c) Given that $Q(x) = x^4 - 5x^3 + 4x^2 + 3x + 9$ has a zero of multiplicity 2, solve the equation $Q(x) = 0$ over the complex field. **3**

(d) Given the function $f(x) = \sqrt{2 - \sqrt{x}}$

(i) What is the domain of $f(x)$? **1**

(ii) Show that $f(x)$ is a decreasing function and deduce the range of $f(x)$. **2**

(iii) By considering the graph of $y = f(x)$, or otherwise, evaluate $\int_0^4 \sqrt{2 - \sqrt{x}} dx$ **3**

Question 8 (15 marks) Use a SEPARATE sheet of paper. **Marks**

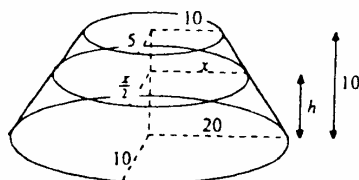
- (a) Consider the rectangular hyperbola $xy = 4$
- (i) Show that the gradient of the tangent at the point $P\left(2p, \frac{2}{p}\right)$ is $-\frac{1}{p^2}$ **1**
- (ii) Show that the equation of the normal at P is given by $p^3x - py = 2(p^4 - 1)$ **1**
- (iii) This normal meets the hyperbola again at $Q\left(2q, \frac{2}{q}\right)$. Prove that $p^3q = -1$. **3**
- (iv) Hence, or otherwise, find the equation of the chord that is a normal at both ends of the chord. **2**

- (b) The line $3y = 5x + 1$ is the equation of the diagonal of a square. One of the square's vertices is $(3, 11)$. Find the coordinates of the other vertices. **3**

- (c) A solid of height 10 metres stands on horizontal ground. The base of the solid is an ellipse with semi-axes 20 metres and 10 metres. Horizontal cross-sections taken parallel to the base and at height h metres above the base are ellipses with semi-axes x metres and $\frac{x}{2}$ metres. The centres of these elliptical cross-sections and the base lie on a vertical straight line, and the extremities of their semi-axes lie on sloping straight lines as shown in the diagram. The top of the solid is an ellipse with semi-axes 10 metres and 5 metres.

Find the volume of the solid correct to the nearest cubic metre.

(you may assume that the area contained by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab). **5**



END OF PAPER

①

Ext 2 Trial 2007 Solutions

$$\begin{aligned}
 a) i) \int \ln x \, dx & \quad \text{let } u = \ln x \quad v = x \\
 & \quad \frac{du}{dx} = \frac{1}{x} \quad \frac{dv}{dx} = 1 \\
 & = x \ln x - \int \frac{1}{x} \, dx \\
 & = x \ln x - \int 1 \, dx \\
 & = x \ln x - x + C
 \end{aligned}$$

$$\begin{aligned}
 i) I &= \int e^x \cos x \, dx \quad \text{let } u = \cos x \quad v = e^x \\
 I &= e^x \cos x - \int e^x \sin x \, dx \quad \frac{du}{dx} = -\sin x \quad \frac{dv}{dx} = e^x \\
 & = e^x \cos x + \int e^x \sin x \, dx \quad u = \sin x \quad v = e^x \\
 & \quad \frac{du}{dx} = \cos x \quad \frac{dv}{dx} = e^x \\
 & = e^x \cos x + e^x \sin x - \int e^x \cos x \, dx \\
 & = e^x (\cos x + \sin x) - I \\
 2I &= e^x (\cos x + \sin x) \\
 I &= \frac{e^x}{2} (\cos x + \sin x)
 \end{aligned}$$

$$j) \text{ let } \frac{4}{4x^2-1} = \frac{4}{(2x-1)(2x+1)} = \frac{A}{2x-1} + \frac{B}{2x+1}$$

$$4 = A(2x+1) + B(2x-1)$$

$$\text{let } x = -1/2, \quad 4 = -2B$$

$$B = -2$$

$$\text{let } x = 1/2, \quad 4 = 2A$$

$$A = 2$$

$$\begin{aligned}
 \therefore \int \frac{4}{4x^2-1} \, dx &= \int \left(\frac{2}{2x-1} - \frac{2}{2x+1} \right) \, dx \\
 &= \ln(2x-1) - \ln(2x+1) + C \\
 &= \ln \left(\frac{2x-1}{2x+1} \right) + C
 \end{aligned}$$

$$\begin{aligned}
 c) \int \frac{dx}{x^2+2x+4} &= \int \frac{dx}{(x+1)^2+3} \\
 &= \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{x+1}{\sqrt{3}} \right)
 \end{aligned}$$

$$\begin{aligned}
 d) \int \sqrt{\frac{x-1}{x+1}} \, dx &= \int \frac{x-1}{\sqrt{x^2-1}} \, dx \\
 &= \frac{1}{2} \int \frac{2x}{\sqrt{x^2-1}} \, dx - \int \frac{dx}{\sqrt{x^2-1}} \\
 &= \sqrt{x^2-1} - \ln(x + \sqrt{x^2-1})
 \end{aligned}$$

$$\begin{aligned}
 Q1e) \int_0^{\pi} \frac{dx}{4 \sin x + 3 \cos x} \\
 &= \int_0^1 \frac{1}{\frac{8t}{1+t^2} + \frac{3(1-t^2)}{1+t^2}} \cdot \frac{2dt}{1+t^2} \\
 &= \int_0^1 \frac{2dt}{3+8t-3t^2} \\
 &= \int_0^1 \frac{2dt}{(3-t)(1+3t)}
 \end{aligned}$$

$$t = \tan\left(\frac{x}{2}\right)$$

$$\sin x = \frac{2t}{1+t^2} \quad \cos x = \frac{1-t^2}{1+t^2}$$

$$\frac{dx}{dt} = \frac{1}{2} \sec^2\left(\frac{x}{2}\right) = \frac{1}{2} \left(1 + \tan^2\left(\frac{x}{2}\right)\right)$$

$$\therefore \frac{dx}{dt} = \frac{2}{1+t^2}$$

$$x=0, t=0 \quad x=\pi, t=1$$

$$\text{let } \frac{2}{(3-t)(1+3t)} = \frac{A}{3-t} + \frac{B}{1+3t}$$

$$\therefore 2 = A(1+3t) + B(3-t)$$

$$\text{let } t=3, \quad 2 = 10A$$

$$A = \frac{1}{5}$$

$$t = -1/3, \quad 2 = 10B$$

$$B = \frac{3}{5}$$

$$\begin{aligned}
 \therefore \int_0^1 \frac{2dt}{(3-t)(1+3t)} &= \int_0^1 \left(\frac{\frac{1}{5}}{(3-t)} + \frac{\frac{3}{5}}{(1+3t)} \right) dt \\
 &= \left[-\frac{1}{5} \ln(3-t) + \frac{1}{5} \ln(1+3t) \right]_0^1 \\
 &= \frac{1}{5} \left[\ln\left(\frac{1+3t}{3-t}\right) \right]_0^1 \\
 &= \frac{1}{5} \left[\ln\left(\frac{4}{2}\right) - \ln\left(\frac{1}{3}\right) \right] \\
 &= \frac{1}{5} \ln 6
 \end{aligned}$$

$$\begin{aligned}
 Q2a) i) \frac{P}{Q} &= \frac{5+2\sqrt{6}i}{1-\sqrt{3}i} \times \frac{1+\sqrt{3}i}{1+\sqrt{3}i} \\
 &= \frac{5+5\sqrt{3}i+2\sqrt{6}i-2\sqrt{18}}{4} \\
 &= \frac{5-6\sqrt{2}}{4} + i \left(\frac{5\sqrt{3}+2\sqrt{6}}{4} \right)
 \end{aligned}$$

$$\begin{aligned}
 ii) \bar{P} \times \bar{Q} &= (5-2\sqrt{6}i)(1+\sqrt{3}i) \\
 &= 5+5\sqrt{3}i-2\sqrt{6}i+2\sqrt{18} \\
 &= 5+6\sqrt{2}+i(5\sqrt{3}-2\sqrt{6})
 \end{aligned}$$

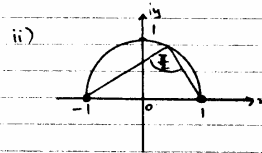
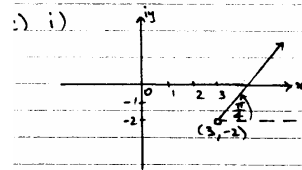
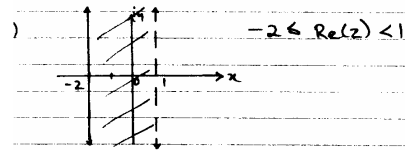
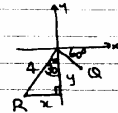
ii) let $\sqrt{P} = x + iy$
 $P = x^2 - y^2 + 2ixy$
 $\therefore x^2 - y^2 = 5$ — (1)
 $2xy = 2\sqrt{6}$ — (2)
from (2) $y = \frac{\sqrt{6}}{x}$ — (3)

Subst (3) into (1) $\Rightarrow x^2 - \frac{6}{x^2} = 5$
 $x^4 - 6 = 5x^2$
 $x^4 - 5x^2 - 6 = 0$
 $(x^2 - 6)(x^2 + 1) = 0$
 $x = \pm\sqrt{6}$
 $y = \pm 1$

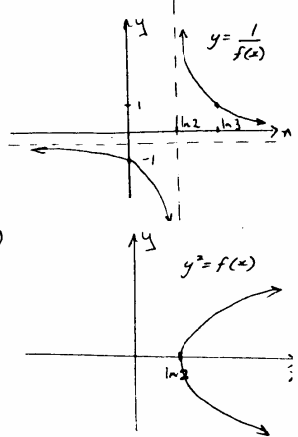
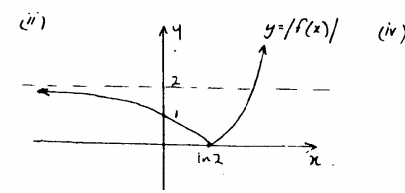
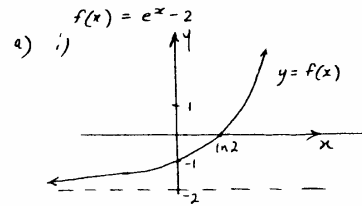
$\therefore \sqrt{P} = \sqrt{6} + i, -\sqrt{6} - i$

i) Modulus: $|Q| = \sqrt{1+9} = 2$, $\arg Q = -60^\circ$

i) $\arg R = -120^\circ$
 $|R| = 4$
 $\therefore R = -2 - 2\sqrt{3}i$

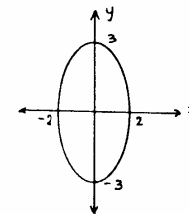


QUESTION 3



b) $V = \lim_{\Delta x \rightarrow 0} \sum_{k=0}^2 (\pi R_k^2 - \pi r_k^2) \Delta x$
 $= \pi \int_0^2 (1 + (x-2)^2 - 1) (1 + (x-2)^2 + 1) dx$
 $= \pi \int_0^2 (x-2)^2 (2 + (x-2)^2) dx$
 $= \pi \int_0^2 2(x-2)^2 + (x-2)^4 dx$
 $= \pi \left[\frac{2(x-2)^3}{3} + \frac{(x-2)^5}{5} \right]_0^2$
 $= \frac{176\pi}{15}$

c) $a^2 = 4$ $b^2 = 9$
 $a = 2$ $b = 3$
 $a^2 = b^2(1 - e^2)$
 $4 = 9(1 - e^2)$
 $e^2 = \frac{5}{9}$
 $e = \frac{\sqrt{5}}{3}$



foci: $S(0, \sqrt{5})$
 $S'(0, -\sqrt{5})$

directrices: $y = \pm 9$

QUESTION 4

$$\begin{aligned} 2) \ i) \ I &= \int \sin^3 x \cdot \cos^5 x \, dx \\ &= \int \sin^3 x (1 - \sin^2 x)^2 \cdot \cos x \, dx \\ &= \int \sin^3 x (1 - 2\sin^2 x + \sin^4 x) \cos x \, dx \\ &= \int (\sin^3 x - 2\sin^5 x + \sin^7 x) \cos x \, dx. \end{aligned}$$

Let $u = \sin x$

$du = \cos x \, dx$

$$\begin{aligned} \therefore I &= \int (u^3 - 2u^5 + u^7) \, du \\ &= \frac{u^4}{4} - \frac{2u^6}{6} + \frac{u^8}{8} \end{aligned}$$

$$\therefore I = \frac{1}{4} \sin^4 x - \frac{1}{3} \sin^6 x + \frac{1}{8} \sin^8 x$$

OR $I = \frac{1}{8} \cos^8 x - \frac{1}{6} \cos^6 x$

Method 2

$$\begin{aligned} I &= \int (1 - \cos^2 x) \cos^5 x \cdot \sin x \, dx \\ &= \int (\cos^5 x - \cos^7 x) \cdot \sin x \, dx \\ &= \int (\cos^7 x - \cos^5 x) \cdot (-\sin x) \, dx \end{aligned}$$

$$= \int u^7 - u^5 \, du$$

$$= \frac{u^8}{8} - \frac{u^6}{6} + c$$

$$= \frac{\cos^8 x}{8} - \frac{\cos^6 x}{6} + c$$

Let $u = \cos x$

$\frac{du}{dx} = -\sin x$

$$ii) \ I = \int \frac{dx}{x^3 \sqrt{x^2 - 4}}$$

Let $x = 2 \sec \theta$

$dx = 2 \sec \theta \tan \theta \, d\theta$

$$\therefore I = \int \frac{2 \sec \theta \tan \theta \, d\theta}{8 \sec^3 \theta \sqrt{4 \sec^2 \theta - 4}}$$

$$= \int \frac{\tan \theta \, d\theta}{4 \sec^2 \theta \cdot 2 \tan \theta}$$

$$= \frac{1}{8} \int \frac{d\theta}{\sec^2 \theta}$$

$$= \frac{1}{8} \int \cos^2 \theta \, d\theta$$

$$= \frac{1}{16} \int (1 + \cos 2\theta) \, d\theta$$

$$= \frac{1}{16} \left(\theta + \frac{1}{2} \sin 2\theta \right)$$

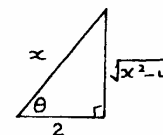
$$= \frac{1}{32} (2\theta + \sin 2\theta) = \frac{1}{32} \left[2 \cos^{-1} \frac{2}{x} + 4 \frac{\sqrt{x^2 - 4}}{x^2} \right] \quad *$$

Since $x = 2 \sec \theta$

$$\frac{x}{2} = \frac{1}{\cos \theta}$$

$$\therefore \cos \theta = \frac{2}{x}$$

$$\theta = \cos^{-1} \left(\frac{2}{x} \right)$$



$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$= 2 \cdot \frac{\sqrt{x^2 - 4}}{x} \cdot \frac{2}{x}$$

$$= \frac{4 \sqrt{x^2 - 4}}{x^2}$$

$$iii) \ I = \int \tan^4 x \, dx$$

$$= \int \tan^2 x \tan^2 x \, dx$$

$$= \int \tan^2 x (\sec^2 x - 1) \, dx$$

$$= \int \sec^2 x \tan^2 x - \tan^2 x \, dx$$

$$= \int \sec^2 x \tan^2 x - \sec^2 x + 1 \, dx$$

$$= \frac{1}{3} \tan^3 x - \tan x + x$$

QUESTION 4

$$1) i) I_n = \int x^n \cos x \, dx$$

$$\text{let } u = x^n \quad dv = \cos x \, dx$$

$$du = nx^{n-1} \, dx \quad v = \sin x$$

$$I_n = x^n \sin x - n \int x^{n-1} \sin x \, dx$$

$$\text{let } u = x^{n-1} \quad dv = \sin x \, dx$$

$$du = (n-1)x^{n-2} \, dx \quad v = -\cos x$$

$$I_n = x^n \sin x - n \left(-x^{n-1} \cos x + \int (n-1)x^{n-2} \cos x \, dx \right)$$

$$= x^n \sin x + nx^{n-1} \cos x - n(n-1) \int x^{n-2} \cos x \, dx$$

$$= x^n \sin x + x^{n-1} \cos x - n(n-1) I_{n-2}$$

$$ii) \text{ let } I_4 = \int x^4 \cos x \, dx$$

$$I_0 = \int \cos x \, dx$$

$$= \sin x$$

$$I_2 = x^2 \sin x + 2x \cos x - 2 \sin x$$

$$I_4 = x^4 \sin x + 4x^3 \cos x - 12(x^2 \sin x + 2x \cos x - 2 \sin x)$$

$$= x^4 \sin x + 4x^3 \cos x - 12x^2 \sin x + 24x \cos x + 24 \sin x$$

$$\int_0^{\pi/2} x^4 \cos x \, dx = \left[x^4 \sin x + 4x^3 \cos x - 12x^2 \sin x + 24x \cos x + 24 \sin x \right]_0^{\pi/2}$$

$$= \left(\frac{\pi}{2} \right)^4 - 12 \left(\frac{\pi}{2} \right)^2 + 24$$

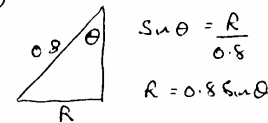
$$= \frac{\pi^4}{16} - 3\pi^2 + 24$$

QUESTION 5

a) i)



ii)



$$T \sin \theta = m \omega^2 R$$

$$T \sin \theta = 3 \times 9\pi^2 \times 0.8 \sin \theta$$

$$T = 21.6\pi^2$$

$$= 213 \, \text{N}$$

$$iii) T \cos \theta = 30$$

$$\cos \theta = \frac{30}{T}$$

$$= 0.14$$

$$\theta = 82^\circ$$

$$b) x = h + \frac{gt}{R} + \frac{ge^{-kt}}{R^2} - \frac{g}{R^2}$$

$$v = \dot{x} = \frac{g}{R} - \frac{ge^{-kt}}{R}$$

$$\dot{x} = ge^{-kt}$$

$$\text{now } v = \frac{g}{R} - \frac{ge^{-kt}}{R}$$

$$kv = g - ge^{-kt}$$

$$ge^{-kt} = g - kv$$

$$\therefore \dot{x} = g - kv$$

$$ii) F = ma$$

$$= mg - mkv$$

gravity and resistance $\propto$ to velocity

$$iii) g = kv_T$$

$$v_T = \frac{g}{R}$$

$$\frac{v_T}{2} = \frac{g}{2R}$$

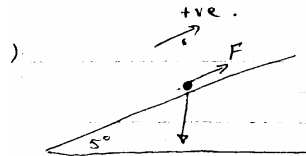
$$\therefore \frac{g}{2R} = \frac{g}{R} - \frac{ge^{-kT}}{R}$$

$$\frac{1}{2} = 1 - e^{-kT}$$

$$e^{-kT} = \frac{1}{2}$$

$$e^{kT} = 2$$

5



$$40 \text{ km/h} = \frac{40000}{3600} \\ = \frac{100}{9} \text{ m/s}$$

$$\frac{d(\frac{1}{2}v^2)}{dx} = a$$

$$\frac{1}{2}v^2 = ax + C$$

$$v^2 = 2ax + 2C$$

$$\text{When } x=0, v = -\frac{100}{9}$$

$$2C = \frac{10000}{81}$$

$$\therefore v^2 = 2ax + \frac{10000}{81}$$

$$\text{When } x = -55, v = 0$$

$$110a = \frac{10000}{81}$$

$$a = 1.12233 \text{ m/s}^2$$

$$\therefore F - mg \sin 5^\circ = 4800 \times 1.12233$$

$$F = 5387.205 + 48000 \sin 5^\circ$$

$$= 9570.68 \text{ N}$$

$$= 9571 \text{ N}$$

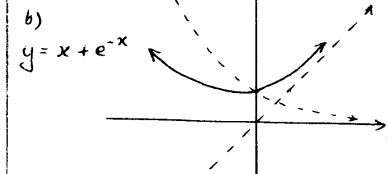
$$\begin{aligned} d) \text{ L.H.S.} &= \frac{\cos y - \cos(y+2\pi)}{2 \sin x} \\ &= \frac{\cos y - \cos y \cos 2\pi + \sin y \sin 2\pi}{2 \sin x} \\ &= \frac{\cos y - \cos y (\cos^2 x - \sin^2 x) + 2 \sin y \cos x \sin x}{2 \sin x} \\ &= \frac{\cos y (1 - \cos^2 x + \sin^2 x) + 2 \sin y \sin x \cos x}{2 \sin x} \\ &= \frac{2 \sin^2 x \cos y + 2 \sin y \sin x \cos x}{2 \sin x} \\ &= \sin x \cos y + \sin y \cos x \\ &= \sin(y+x) \\ &= \text{R.H.S.} \end{aligned}$$

QUESTION 6

$$\begin{aligned} a) i) \Delta V &= \pi(R^2 - r^2)h \\ &= \pi(R-r)(R+r)h \\ &= \pi(3-x-3+x+\Delta x)(3-x+3-x-\Delta x)h \\ &= \frac{6\pi}{\sqrt{4-x^2}} (6-2x-\Delta x) \Delta x \\ &= \frac{6\pi}{\sqrt{4-x^2}} (6-2x) \Delta x \end{aligned}$$

$$\begin{aligned} \therefore V &= \lim_{\Delta x \rightarrow 0} \sum_{i=0}^n \Delta V \\ &= 6\pi \int_0^1 \frac{6-2x}{\sqrt{4-x^2}} dx \\ &= 12\pi \int_0^1 \frac{3-x}{\sqrt{4-x^2}} dx \end{aligned}$$

$$\begin{aligned} ii) V &= 12\pi \left(\int_0^1 \frac{3 dx}{\sqrt{4-x^2}} + \frac{1}{2} \int_0^1 \frac{-2x dx}{\sqrt{4-x^2}} \right) \\ &= 12\pi \left[3 \sin^{-1} \frac{x}{2} + \sqrt{4-x^2} \right]_0^1 \\ &= 4\pi \end{aligned}$$



turning point (0, 1)
asymptote $y = x$

$$\begin{aligned} \frac{dy}{dx} &= 1 - e^{-x} = 0 \\ e^{-x} &= 1 \\ x &= 0 \end{aligned}$$

$\therefore$ t.p. when $x=0, y=1$.

$$\begin{aligned} ii) \Delta V &= \pi(R^2 - r^2)h \\ &= \pi((x+\Delta x)^2 - x^2)(x+e^{-x}-x) \\ &= \pi(x^2 + 2x\Delta x + \Delta x^2 - x^2)e^{-x} \\ &= \pi(2x\Delta x + \Delta x^2)e^{-x} \\ \therefore V &= 2\pi \int_0^1 x e^{-x} dx \\ \text{Let } u &= x \quad dv = e^{-x} dx \\ du &= dx \quad v = -e^{-x} \\ \therefore V &= 2\pi \left[-x e^{-x} + \int e^{-x} dx \right]_0^1 \\ &= 2\pi \left[-x e^{-x} - e^{-x} \right]_0^1 \\ &= 2\pi (-e^{-1} - e^{-1} - 0 + 1) \\ &= 2\pi \left(1 - \frac{2}{e} \right) \\ &= 1.66 \end{aligned}$$

$$\begin{aligned} c) 12 + L &= L^2 \\ L^2 - L - 12 &= 0 \\ (L-4)(L+3) &= 0 \\ \therefore L &= -3, 4 \\ \therefore L &= 4 \text{ as } L > 0 \end{aligned}$$

QUESTION 7

Let roots be $\alpha, \alpha k, \alpha k^2$

$$\therefore \alpha(1+k+k^2) = \frac{-q}{p} \quad \text{--- (1)}$$

$$\alpha^2 k(1+k+k^2) = \frac{r}{p} \quad \text{--- (2)}$$

$$\alpha^3 k^3 = \frac{-s}{p} \quad \text{--- (3)}$$

$$\therefore \text{dividing (1) by (3)} \Rightarrow \alpha k = \frac{-r}{-s} \quad \text{--- (4)}$$

dividing (2) by (4)

$$\text{to (3)} \Rightarrow \frac{-r^3}{s^3} = \frac{-s}{p}$$

$$pr^3 = q^3 s$$

$$1) (z-1)(z+1) = z^2 + 1$$

$$\frac{z^2 + 2z - 3}{z^2 + 1} = \frac{z^2 + 2z - 3}{z^2 + 1}$$

$$\text{Now } z^2 + 2z - 3 = (z+3)(z-1)$$

$\therefore$ other roots are $-1, 1, -3$

$$2) Q(x) = x^4 - 5x^3 + 4x^2 + 3x + 9$$

$$Q'(x) = 4x^3 - 15x^2 + 8x + 3$$

$$Q'(3) = 0$$

$$Q(3) = 0$$

$x = 3$ double root

$$(x-3)^2 = x^2 - 6x + 9$$

$$\begin{array}{r} x^2 + x + 1 \\ (x^2 - 6x + 9) \overline{) x^4 - 5x^3 + 4x^2 + 3x + 9} \end{array}$$

$$\therefore x = 3, -\frac{1}{2} \pm i\frac{\sqrt{3}}{2}$$

$$1) i) 0 \leq x \leq 4$$

$$ii) f'(x) = \frac{-1}{4\sqrt{x(2-\sqrt{x})}} < 0$$

$$R: 0 \leq y \leq \sqrt{x}$$

$$iii) y = \sqrt{2-\sqrt{x}} \\ x = (2-y^2)^2$$

$$A = \int_0^{\sqrt{2}} 4 - 4y^2 + y^4 dy \\ = \left[4y - \frac{4y^3}{3} + \frac{y^5}{5} \right]_0^{\sqrt{2}}$$

$$= \frac{32\sqrt{2}}{15}$$

$$* 8c) x = mh + b$$

$$\text{when } h=0, x=20$$

$$\therefore b=20$$

$$\text{when } h=10, x=10$$

$$\therefore 10 = 10h + 20$$

$$h = -1$$

$$\therefore x = 20 - h$$

$$A = \pi x \times \frac{x}{2}$$

$$= \frac{\pi}{2} (20-h)^2$$

$$V = \frac{\pi}{2} \int_0^{10} (20-h)^2 dh$$

$$= \frac{\pi}{6} \left[(20-h)^3 \right]_0^{10} \\ = 3665 \pi \text{ m}^3$$

QUESTION 8

$$a) i) xy = 4 \\ y = 4x^{-1}$$

$$\frac{dy}{dx} = \frac{-4}{x^2}$$

$$\text{when } x = 2p$$

$$\frac{dy}{dx} = \frac{-4}{4p^2}$$

$$= \frac{-1}{p^2}$$

$$ii) y - \frac{2}{p} = p^2(x-2p)$$

$$yp - 2 = p^3x - 2p^3$$

$$p^3x - py = 2p^3 - 2$$

$$p^3x - py = 2(p^3 - 1)$$

$$iii) \text{subst } (2p, \frac{2}{p}) \text{ into normal}$$

$$p^3(2p) - \frac{2p}{p} = 2(p^3 - 1)$$

$$2p^4 - 2 = 2p^3 - 2$$

$$2p^4 - 2p^3 = 2p - 2$$

$$2p^3(p - 1) = 2(p - 1)$$

$$p^3 = \frac{p-1}{p-1}$$

$$p^3 = -1$$

$$\text{OR } p^3x - py = 2(p^3 - 1) \quad \text{--- (1)}$$

$$y = \frac{p^3x - 2(p^3 - 1)}{p} \quad \text{--- (2)}$$

$$\text{subst (1) into (2)} \Rightarrow p^3x - \frac{p^3x - 2(p^3 - 1)}{p} = 2(p^3 - 1)$$

$$p^3x^2 - 4p = 2p^3x - 2p$$

$$p^3x^2 + x(2 - 2p^3) - 4p = 0$$

now product of roots

$$2p \times 2q = \frac{-4p}{p^3} \\ pq = -\frac{1}{p^2}$$

$$q = \frac{-1}{p^3} \\ p^3q = -1$$

iv) to be normal at $P+Q$

$$p^3 = q^2 \therefore q = \pm p$$

subst into $p^3q = -1$

$$\therefore p^4 = -1 \text{ and } p^4 = 1$$

$$\therefore p = \pm 1$$

subst into eqn of normal

$$\text{when } p=1, p^3x - py = 2(p^3 - 1)$$

$$x - y = 0$$

$$y = x$$

$$\text{when } p=-1, p^3x - py = 2(p^3 - 1)$$

$$-x + y = 0$$

$$y = x$$

$$\therefore y = x$$

$$b) y = \frac{3x}{5} + \frac{1}{3} \quad \text{--- (1)}$$

grad other diagonal = $-\frac{3}{5}$

$$\therefore \text{eqn diag} \Rightarrow y = -\frac{3x}{5} + b$$

subst (3,1) into eqn of diag

$$1 = -\frac{9}{5} + b$$

$$b = \frac{64}{5}$$

$$\therefore y = -\frac{3x}{5} + \frac{64}{5} \quad \text{--- (2)}$$

solving (1) & (2) simultaneously

$$\frac{3x}{5} + \frac{1}{3} = -\frac{3x}{5} + \frac{64}{5}$$

$$25x + 5 = -9x + 192$$

$$x = 5\frac{1}{2}$$

$$y = 9\frac{1}{2}$$

} middle of square

by symmetry (8,8) (7,12) (4,7)

* 8c)